

Build an Offline ChatGPT-Like Tool

This comprehensive DIY guide offers a quick way to build an offline, on-premises ChatGPT-like tool using Ollama and Ollama-WebUI.

ChatGPT has been widely adopted across various sectors because it can understand and generate human-like text. Its applications range from customer service automation, where it can handle enquiries and provide support 24/7, to content creation, aiding in drafting articles, emails, and reports. Its integration into educational tools facilitates personalised learning experiences, and it assists developers by debugging code and generating programming-related content.

Open source AI models

Open source models such as META's LLaMA2 and Microsoft's Phi2 offer a foundation for building customised AI solutions, democratising access to cutting-edge technology. The development of offline AI solutions, particularly those based on open source projects like Ollama and Ollama-WebUI, marks a significant step forward in AI. These tools not only provide the benefits of AI models like ChatGPT but also ensure that organisations can innovate and maintain control over their data, comply with privacy regulations, and operate without the need for constant internet connectivity.

This guide will walk you through the fastest DIY method to set up an offline, on-prem ChatGPT-like tool using Ollama and Ollama-WebUI.



Setting up your OS environment

Before diving into the installation of your offline, on-prem ChatGPT tool, it's crucial to ensure that your environment meets the necessary system requirements and that Debian Linux 12 is correctly installed on your machine, whether it's a physical server or a virtual machine (VM).

To ensure smooth operation of your offline AI solution, your system should meet the following minimum specifications.

- **CPU:** Latest model (Intel i5, Xeon, or AMD EPYC) with at least 8 cores
- **RAM:** 16GB to start with or more
- **HDD storage:** 100GB or more
- **NVIDIA GPU:** Optional (but recommended for enhanced performance)
Here's a step-by-step guide to installing Debian.
Download the Debian installation image: Visit the official Debian website at <https://cdimage.debian.org/debian-cd/current/amd64/iso-cd/debian-12.4.0-amd64-netinst.iso> to download the latest Debian installation image suitable for your system architecture.
Create a bootable USB drive or CD/DVD: Use software like Rufus or any other preferred tool to create a bootable USB drive or burn the Debian image onto a CD/DVD.